

Chih-Wei Joshua Liu

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EDUCATION

Stanford University

Stanford, CA, USA

Biophysics PhD Program · **Thesis advisor:** Grant M. Rotskoff · **GPA:** 4.0/4.0

09/2024 – present

Courses: IMMUNOL201 Advanced Immunology, SBIO242 Molecular Biophysics Methods, CS229 Machine Learning

Massachusetts Institute of Technology (MIT)

Cambridge, MA, USA

Bachelor of Science · **Major:** Physics · **GPA:** 4.9/5.0 · **MCAT:** 528/528

05/2022

Courses: 18.650 Statistics, 8.591 Systems Biology · **Honors:** Joel Matthew Orloff Award, [MIT News profile](#)

National Experimental High School

Hsinchu, Taiwan, ROC

USABO: Semifinalist (2015 – 2018); High Honorist (2017) · **ACT:** 36/36 · **SAT:** 1600/1600

05/2018

EXPERIENCE

MIT Physics of Living Systems Group

06/2022 – 08/2024

Research Assistant

Cambridge, MA, USA

Implemented microscopy experiments. Designed ensemble-learning and polyspectral analyses of biological images.

MIT Department of Physics

02/2024 – 05/2024

Teaching Assistant, 8.241 Introduction to Biological Physics

Cambridge, MA, USA

Led recitations on protein folding, diffusion, and imaging methods. Assisted optical-trap experiments.

Massachusetts General Hospital

06/2021 – 08/2021

Data Science Intern

Boston, MA, USA

Implemented GLS models of patient data and authored two clinical publications using R, SPSS, and `statsmodels`.

SELECTED PROJECTS

C.-W.J. Liu, J. Klinger, G.M. Rotskoff, Optimal parameterization of nonequilibrium generalized master equations from discrete-time experimental data, [arXiv:2606.28289](#), in revision at *Phys. Rev. X* (2026). [\[code\]](#)

- Derived the first maximum-likelihood parameterization of non-Markovian kinetics satisfying all physical constraints in general biomolecules. **Skills:** convex optimization, `lightning`
- Validated algorithm on CFTR ion-channel and F1 molecular-motor experiments. Set SOTA in benchmark extraction of HP35 protein-folding kinetics from molecular-dynamics simulations. **Skills:** autoregressive models, `mdtraj`

J. Li*, C.-W.J. Liu*, M. Szurek, and N. Fakhri, Measuring irreversibility from learned representations of biological patterns, *PRX Life* **2**, 033013 (2024), *equal contribution. [\[code\]](#)

- Quantified energetic costs of actomyosin regulation by conserved Rho GTPase using only variational-autoencoder representations of cortical-wave microscopy data. **Skills:** `pillow`, `torch`, `torchvision`

A. Yona, C.-W.J. Liu, et al., and J. Gore, Adaptive evolution identifies candidate genetic routes associated with oxalate utilization in probiotic and gut-derived *E. coli*, in revision at *Cell Rep. Methods* (2026).

- Led biochemical characterization of human gut microbes subject to directed evolution, identifying loci repeatedly mutated across lineages with sequencing and imaging assays. **Skills:** metagenomics, anaerobic technique

H. Lee*, E. Koskelo*, S. Gokhale*, J. Li*, C. Fei, C.-W.J. Liu, L. Lin, J. Dunkel, D.J. Skinner, and N. Fakhri, Topological flowscape reveals state transitions in nonreciprocal living matter, *Phys. Rev. X* **16**, 031061 (2026), *equal contribution. Featured in *Physics* magazine and on the cover of *Phys. Rev. X*.

- Trained classifier of developmental stages in embryo videos, enabling identification of physically exotic swarming dynamics in multicellular aggregates. **Skills:** `scikit-learn`, `scipy`, `numpy`, `pandas`, `opencv`

GRANTS AND LEADERSHIP

National Science Foundation Graduate Research Fellowship (GRFP, 2024, Physics)

National Defense Science and Engineering Graduate Fellowship (2024, Physics, declined for GRFP)

MIT Society of Physics Students and Sigma Pi Sigma: honor-society inductee (2022); President (2022)

Phi Kappa Theta – Massachusetts Eta: Pledgemaster (2021); sourced and fundraised donation of 12,000 surgical masks to Beth Israel Deaconess Medical Center during pandemic shortage as Service Chair (2019–2020).

SKILLS AND INTERESTS

Biology: cloning, enzyme assays, fluorescence microscopy · **Computing:** `jax`, `pydantic`, `uv`, `ty`, `ruff`, `slurm`

Natural language: Standard Chinese (native) · **Personal interests:** phonology, archaeology, ichthyology, Set

Research interests: nonequilibrium thermodynamics, biochemical kinetics, optimal transport, SAT solvers